

Machine Learning 1

Mock Homework-based Midterm 2 — A

Two hours 18 points Two exercises

Name: _____

Date: _____

Instructions

- This is an unofficial, deliberately challenging practice paper, not a prediction of the actual exam.
- Show your reasoning and intermediate steps. Partial credit is available. An unsupported final answer is not sufficient.
- For multivariate derivatives, first derive an arbitrary component in index notation, then assemble the vector or matrix.
- Use numerator layout: the Jacobian of $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ has shape $n \times m$. We write $\nabla_w f$ for the column gradient, so $df/dw = (\nabla_w f)^\top$ for scalar f .
- All vectors are columns. All logarithms are natural. Treat fitted parameters as fixed when making predictions.
- Use a basic calculator if needed. Exact expressions are acceptable. Do not use the answer key while attempting the paper.
- Suggested timing: 65 minutes for Exercise 1, 45 minutes for Exercise 2, and 10 minutes to check your work.

Exercise	Topic	Points
1	Generative classification and regularized softmax	10
2	Affine PCA, a constrained component, and PPCA	8

Useful definitions

$$\sigma(a) = \frac{1}{1 + e^{-a}}, \quad \text{softmax}(a)_k = \frac{e^{a_k}}{\sum_j e^{a_j}}.$$

For symmetric positive definite $C \in \mathbb{R}^{D \times D}$,

$$\mathcal{N}(x \mid \mu, C) = \frac{\exp[-\frac{1}{2}(x - \mu)^\top C^{-1}(x - \mu)]}{(2\pi)^{D/2} |C|^{1/2}}.$$

Exercise 1: Two sensors, one class-dependent rate

10 points

A device belongs to one of K classes. Two sensors give an input $x = (m, s)^\top$, where $m \in \{0, 1, \dots\}$ is a count and $s \geq 0$ is a waiting time. The sensors are independent *conditional on the class*. For class C_k ,

$$p(m \mid C_k) = \frac{(2\lambda_k)^m e^{-2\lambda_k}}{m!}, \quad p(s \mid C_k) = \lambda_k e^{-\lambda_k s}, \quad \lambda_k > 0.$$

The same λ_k occurs in both sensors; it is not two separate parameters. The class priors $\pi_k > 0$, with $\sum_k \pi_k = 1$, are also unknown. The training observations (m_n, s_n, t_n) are independent, and t_n is a one-hot class vector. Define

$$N_k = \sum_n t_{nk}, \quad M_k = \sum_n t_{nk} m_n, \quad S_k = \sum_n t_{nk} s_n.$$

Assume $N_k > 0$ for every class.

1a [3 p] Write the joint likelihood of the observed inputs and labels, then its log-likelihood. Derive the maximum-likelihood estimates of *both* π_k and λ_k . Enforce the prior-probability constraint with a Lagrange multiplier and verify that the rate estimate is a maximum.

Exercise 1 continued

For the remaining generative-model questions, use this training summary:

Class	N_k	M_k	S_k
C_1	4	12	8
C_2	2	14	4
C_3	2	2	4

1b [2 p] Calculate the fitted priors and rates. Express the posterior probabilities as a softmax of scores affine in (m, s) . Give the two explicit inequalities that define the region in which C_1 has posterior probability at least as large as each competitor. If m is temporarily allowed to be a nonnegative real number, is this region convex? Justify your answer.

1c [1 p] For the complete input $(m, s) = (0, \frac{1}{2})$, identify the highest-posterior class. Now suppose the time sensor fails independently of the input and class, so only $m = 0$ is observed. Derive the correct posterior from the count alone and identify its highest-posterior class. Explain why setting the missing waiting time to zero is not marginalization.

Exercise 1 continued

1d [1 p] Generate a new device by sampling its class from the fitted priors, then sample its two sensors from that class. You may use $\mathbb{E}[M \mid C_k] = 2\lambda_k$ and $\mathbb{E}[S \mid C_k] = 1/\lambda_k$. Derive and calculate the *unconditional* covariance $\text{Cov}(M, S)$. Does conditional independence imply unconditional independence here?

Next, fit a *discriminative* classifier to the original labeled data, using $\phi_n = (1, m_n, s_n)^\top$ and freely adjustable $w_k \in \mathbb{R}^3$:

$$y_{nk} = \frac{\exp(w_k^\top \phi_n)}{\sum_j \exp(w_j^\top \phi_n)}, \quad p(w_1, \dots, w_K) = \prod_k \mathcal{N}(w_k \mid 0, (\alpha R)^{-1}).$$

Here $\alpha > 0$ and $R \in \mathbb{R}^{3 \times 3}$ is a known symmetric positive definite matrix. The prior includes the intercepts.

1e [2 p] Obtain a negative log-posterior objective, omitting constants independent of the weights. Derive $\partial L / \partial w_{jr}$ in index notation and assemble $\nabla_{w_j} L$. Do not assume that R is diagonal.

1f [1 p] Without the prior, show that replacing every w_k by $w_k + v$, for the same $v \in \mathbb{R}^3$, leaves the predictions unchanged. Prove that the MAP minimizer under the stated prior satisfies $\sum_k w_k = 0$. Explain what the prior does to this parameter ambiguity.

Exercise 2: PCA with an offset and a forbidden direction

8 points

A dataset contains J distinct vectors $x_i \in \mathbb{R}^3$. Vector x_i occurs $n_i > 0$ times among the $N = \sum_i n_i$ observations. Set $q_i = n_i/N$, so $\sum_i q_i = 1$. The full dataset has mean and covariance

$$\bar{x} = \sum_i q_i x_i = \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}, \quad S = \sum_i q_i (x_i - \bar{x})(x_i - \bar{x})^\top = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The covariance uses the $1/N$, not the $1/(N-1)$, convention. A one-dimensional *affine* reconstruction uses a unit vector u and an offset c :

$$\hat{x}_i = c + uu^\top(x_i - c), \quad E(c, u) = \sum_i q_i \|x_i - \hat{x}_i\|^2, \quad u^\top u = 1.$$

2a [2 p] For a fixed u , derive *all* minimizing offsets c . Show how the remaining minimization reduces to a trace expression involving S and $uu^\top$. Explain why taking $c = \bar{x}$ is valid but does not make the offset unique.

2b [1 p] With $c = \bar{x}$, find an optimal unconstrained unit vector u , the retained variance, and the minimum average reconstruction error. For the new input $x_* = (6, -1, 3)^\top$, give its scalar coordinate $z_* = u^\top(x_* - \bar{x})$, its reconstruction in $\mathbb{R}^3$, and its squared reconstruction error.

Exercise 2 continued

2c [2 p] A measurement policy now forbids the direction $v = (1, 1, 0)^\top / \sqrt{2}$: require $u^\top v = 0$ as well as $u^\top u = 1$. Use a Lagrangian to find all stationary directions of the retained variance. Select the global maximum and give the resulting average reconstruction error. Also reconstruct x_* and calculate its squared error under this constraint.

For the next questions, remove the direction restriction and fit standard PPCA to the full dataset of N observations:

$$z \sim \mathcal{N}(0, 1), \quad x = wz + \mu + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \tau I_3),$$

with independent z, ϵ and $\tau > 0$. Fix $\mu = \bar{x}$ and constrain $w = \sqrt{a} u_1$, where $a \geq 0$ and u_1 is the leading eigenvector of S . The marginal covariance is $C = au_1 u_1^\top + \tau I_3$.

2d [2 p] Write the marginal log-likelihood as a function of a and τ , up to parameter-independent constants, using the eigenvalues of S . Derive its maximizing values of a and τ ; do not only quote the PPCA formula. Give the fitted C explicitly. Does it equal S ? Explain the discrepancy in the discarded directions.

Exercise 2 continued

2e [1 p] An alternative parameterization uses $z' \sim \mathcal{N}(2, 9)$ and $x = w'z' + \mu' + \epsilon$, with the *same fitted noise variance* as in 2d. Find w' and μ' that preserve the fitted marginal distribution of x . Could this new latent prior make a one-dimensional PPCA model match S exactly? Justify your answer using covariance rank or eigenvalues.

Additional working space

Label any answer continued here with its question number.

End of Mock Midterm 2 — A